Section 2.2 — Polynomial Arithmetic — Answer Key

Math 103: College Algebra

Adding Polynomials

Add. Simplify your answer as much as possible.

1. $7m - 3$
2. $6w^2 - 10w$
3. $2r^2 + 1.4r - 7.8$
4. $2a^3 + 3a^2 + \frac{7}{3}a - \frac{1}{12}$
5. $-\frac{1}{15}axy + \frac{20}{21}ax - \frac{16}{35}ay$
6. $x^4 + \frac{1}{3}x^3 + \frac{2}{3}x^2 - 2x + 11$
7. $-2x^2y^2 + 10xy^2 - 8x^2y - 5xy + 13y^2 - 5$
8. $-8w^4 + 3w^3z - 7w^2z^2 + 13wz^3 - 9z^4$

Negating Polynomials

Simplify.

9. $-2m - 3$
10. $-8y - 5$
11. $5y^2 - \frac{1}{5}y + 8$
12. $2x^4 + 3x^2 + 5$

Negation and Subtraction

Simplify. Rewrite as subtraction.

13. $-x + 2 = 2 - x$
14. $-x^2 + y^2 = y^2 - x^2$
15. $-6 + 2x = 2x - 6$
16. $-4 + \frac{1}{2}z = \frac{1}{2}z - 4$
17. Negating the binomial reverses the order of the subtraction.

Subtracting Polynomials

Subtract. Simplify your answer as much as possible.

18. $7b^2 + 3b$
19. $2y^2 - 3y + 8$
20. $5y^2 + y + 16$
21. $3u^2v - \frac{25}{3}uv^2 - 19v^3$
22. $-4x$
23. $-4s^4 + 7s^3 + 2s^2 - 12s - 6$
24. $3.8m^3n + 1.6m^2n^2 + 7.3mn^3$

Multiplying by a Monomial

Multiply. Simplify your answer as much as possible.

25. $28n^2 + 8n$
26. $-0.4b^4 + 1.2b^2c^2$
27. $\frac{2}{3}x^2y^3 - xy^5$
28. $-15hk^6 + 10h^3k^4 + 15h^2k^7 - 20h^2k^3$
29. $-20a^2b^4 + 12ab^7 - 8a^2b^6$

Multiplying Binomials

Multiply. Simplify your answer as much as possible.

30. $a^2 + 13a + 36$
31. $8x^2 + 22x + 12$
32. $14y^2 + 34xy + 12x^2$
33. $a^2 - b^2$
34. $2x^2y^4 - 4x^2y^3z - 2x^3y^3 + 4x^3y^2z$
35. $4x^2 + 12x + 9$

Multiplying Polynomials

Multiply. Simplify your answer as much as possible.

36. $6x^3 + 18x^2 - x - 15$

37. $u^4 + u^3 - 2y^2 - 3y - 1$

38. $x^3 + 64$

39. $a^6 - 2a^5 + 13a^4 + 6a^3 + 18a^2 + 108a + 81$

40. $x^3 + 9x^2 + 27x + 27$

41. $x^3 - 2x^2 - 5x - 6$

42. $4x^3 - 16x^2 + 13x - 3$

43. $-2x^3 + 5x^2 + 17x - 20$

44. $10z^4 - 15z^3 + 31z^2 + 6z - 14$